Self Distillation on Graphs – Minimizing the Drawbacks of Deep Layers

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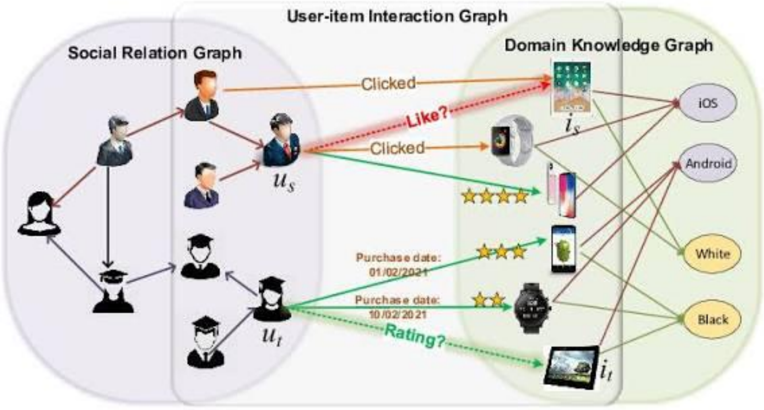
- ☐ SAIL experiments

Use of Graphs

- ☐ Graphs use **relationship information** for various purposes

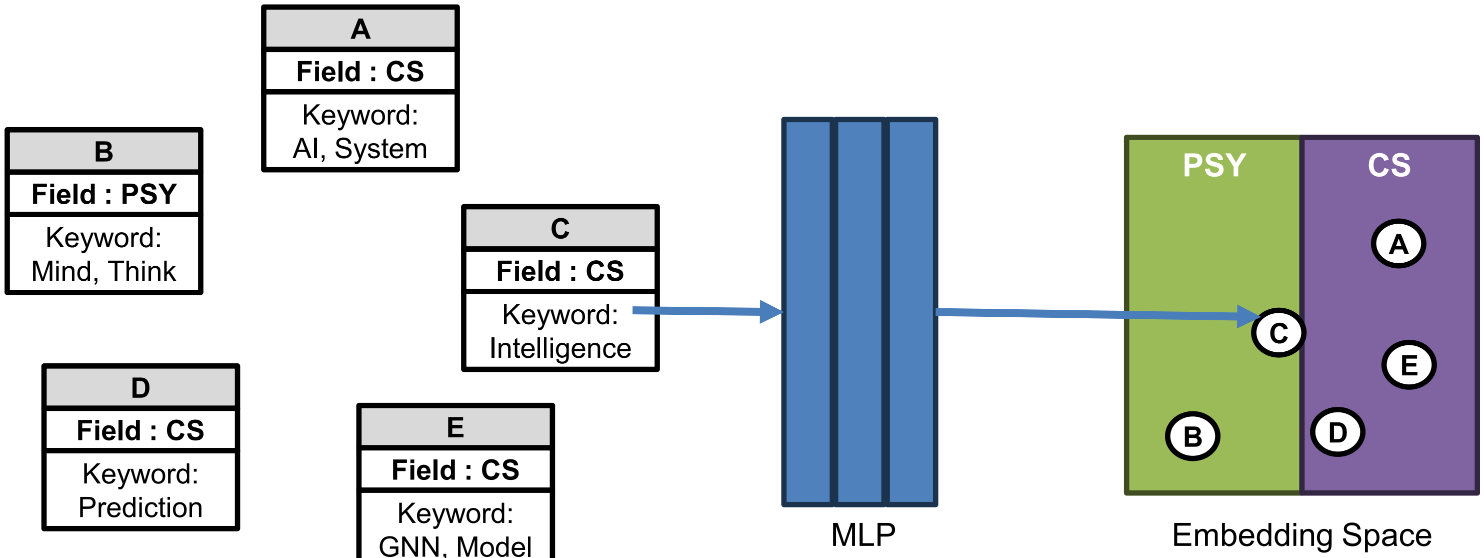
Social Graphs



User-Item Graphs

Limitation of MLP

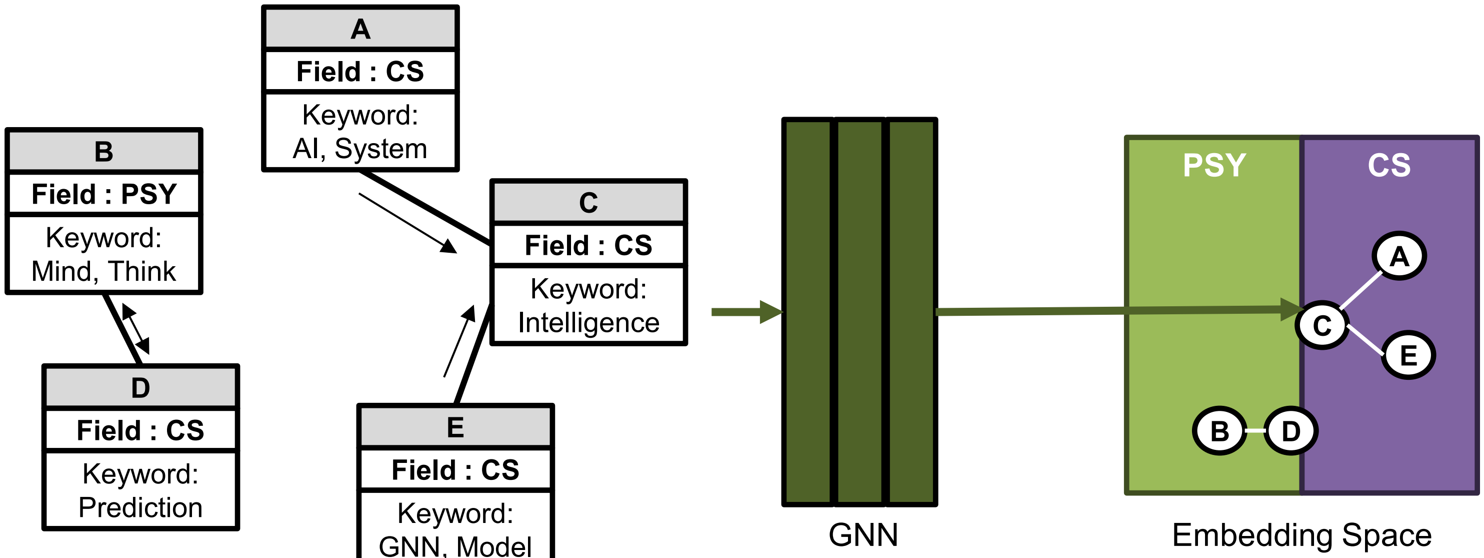
- If features on **individual** nodes are **insufficient**, inference might go **wrong**



Co-Citation Paper Author Field Prediction

Graph Neural Network (GNN)

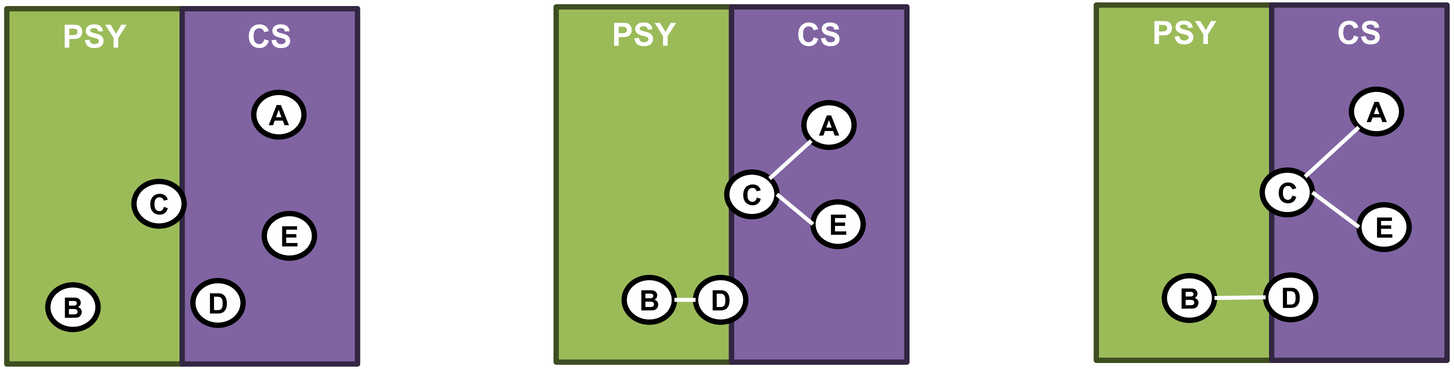
- GNN **aggregates** information of **connected nodes** to use for prediction



Co-Citation Paper Author Field Prediction

Over-Smoothing of GNN

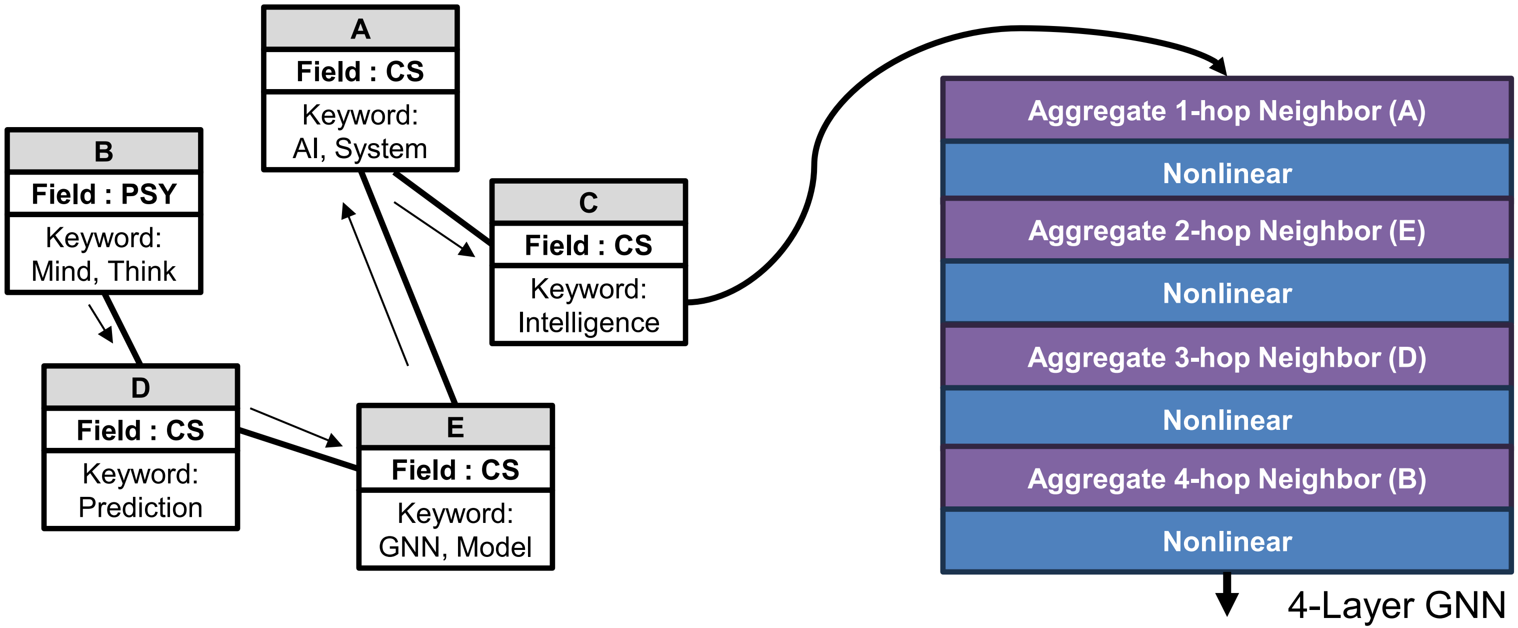
- Too **much smoothing** might **lose** useful **individual information**
  - The embeddings of the nodes **converge** to a particular **vector frequency**



Case of MLP   Case of GNN   What We Wanted

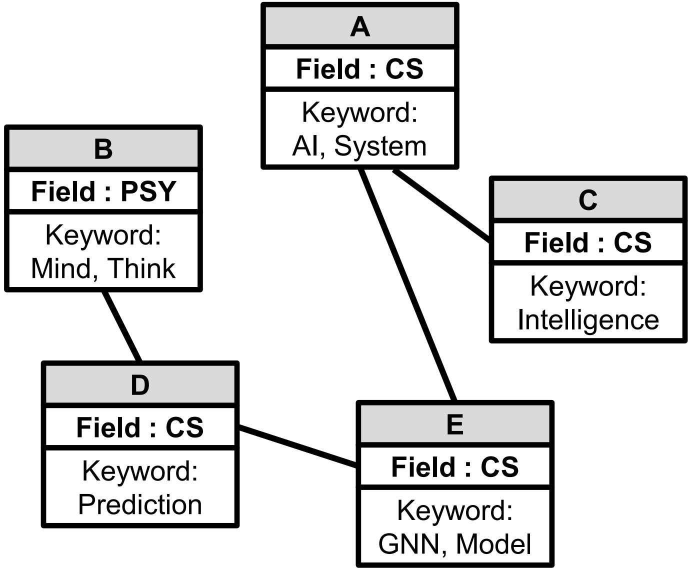
Deep Layer in GNN

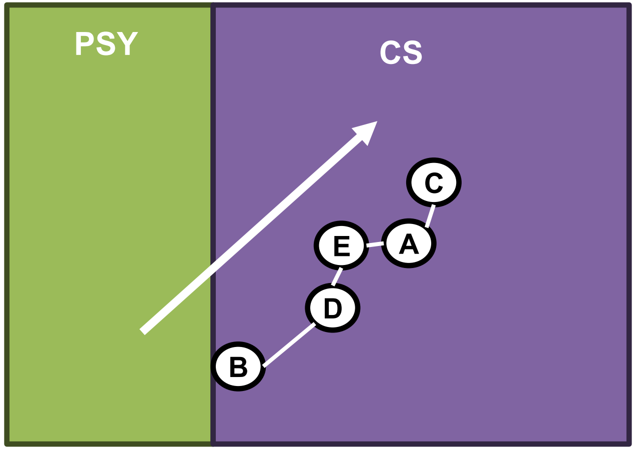
- Using **deep layers** in GNN helps **reach further** relevant hops



Over-Smoothing in Deep Layer in GNN

- **Deep layer** structures in GNN are fragile to **over-smoothing**





Embedding Space

On Self-Distilling Graph Neural Network

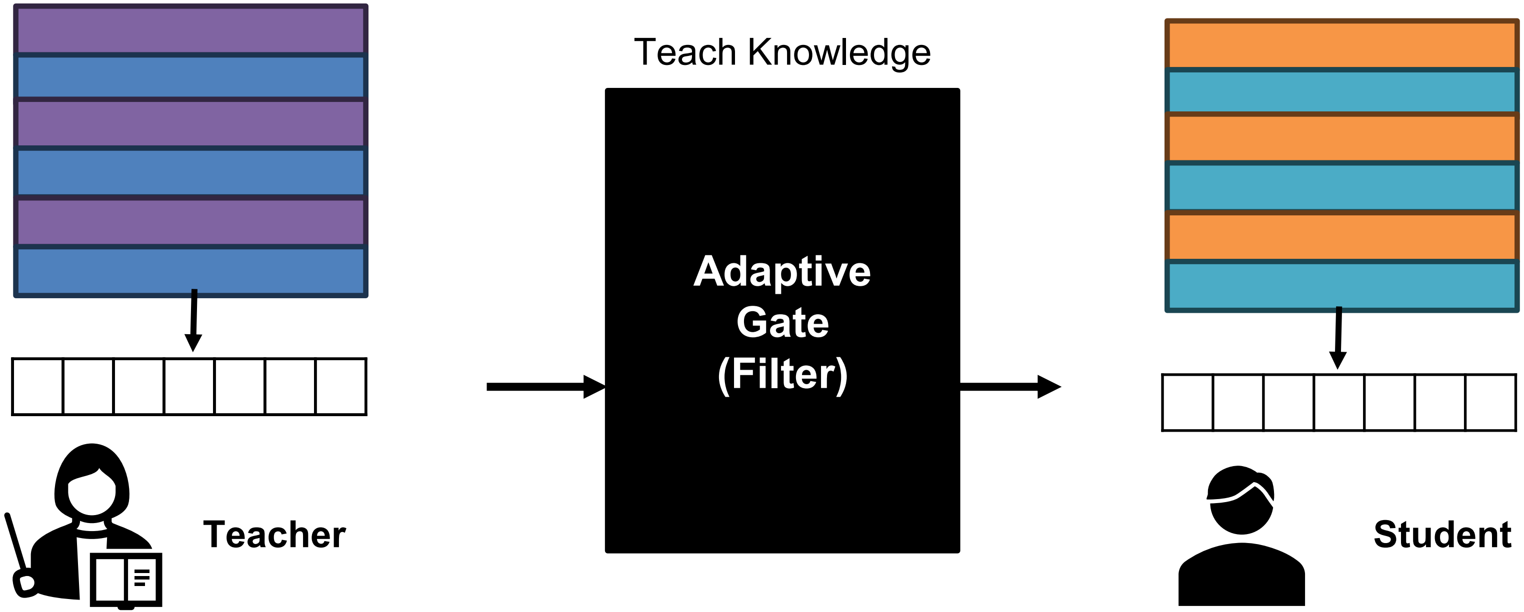
IJCAI 21

Yuzhao Chen[1,2,*], Yatao Bian[2], Xi Xiao[1,3], Yu Rong[2], Tingyang Xu[2], Junzhou Huang[2,4]

[1]Tsinghua University, [2]Tencent AI Lab, [3]Peng Cheng Laboratory, [4]University of Texas at Arlington

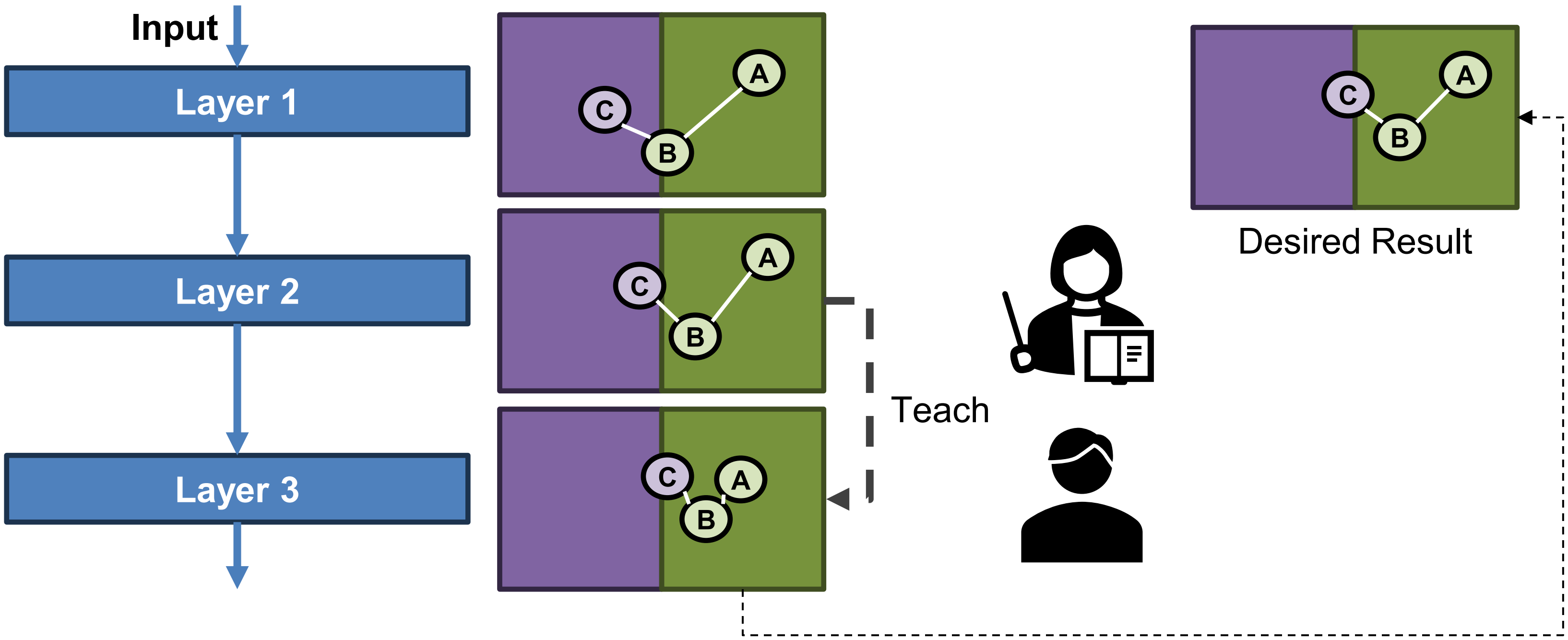
What is Distillation?

- A teacher **teaches useful knowledge** to the student
  - Knowledge to teach is logits, embeddings etc.



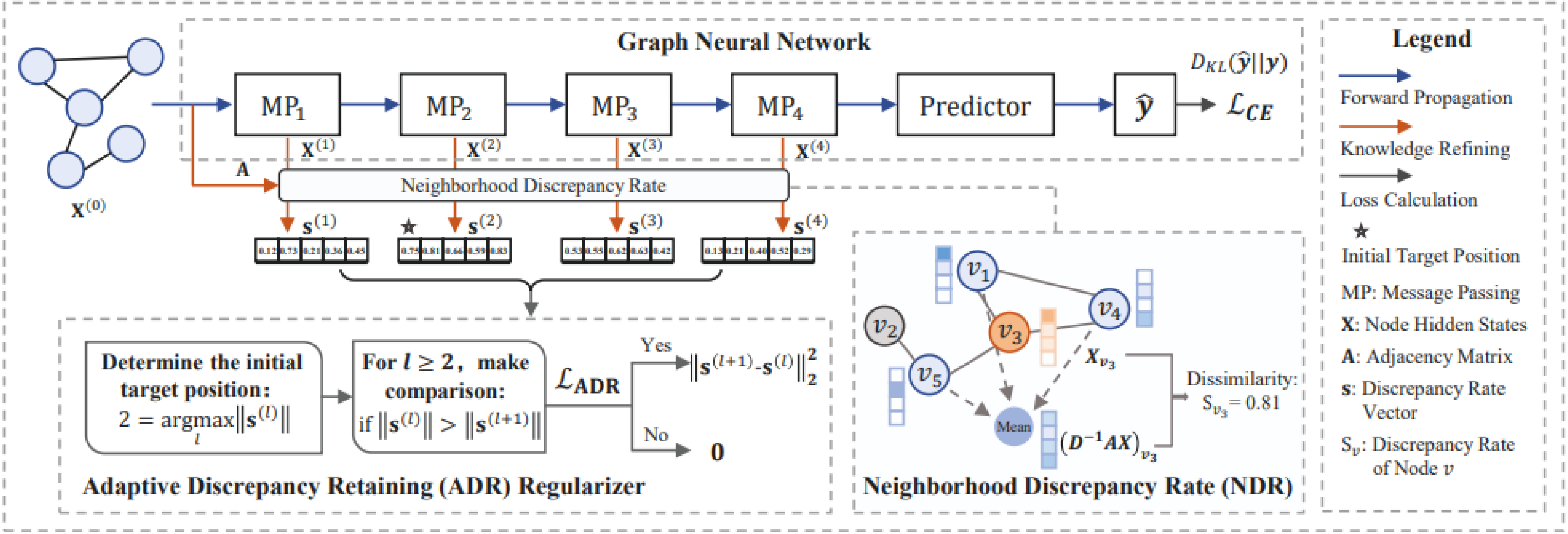
Self-Distillation Between Layers

- ☐ Distillation **between layers** helps **prevent over-smoothing**



GNN-SD Framework

- Check **if over-smoothing happened** and **fix** it by distillation



Neighborhood Discrepancy Rate (NDR)

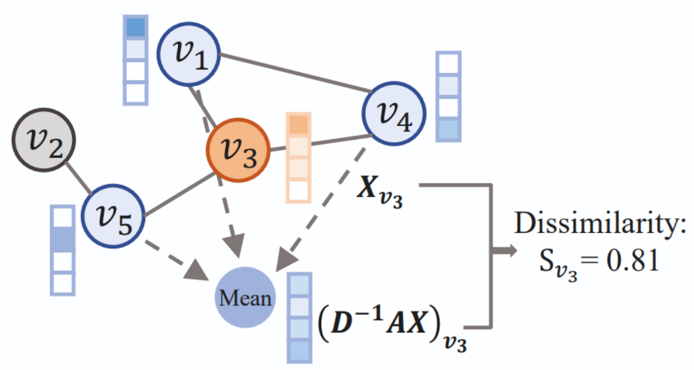
☐ **Quantify the non-smoothness** of each GNN Layer

$$S_v^{(l)} = 1 - \frac{\mathbf{X}_{v\cdot}^{(l)} (\mathbf{A}\mathbf{X}^{(l)})_{v\cdot}^{\mathrm{T}}}{||\mathbf{X}_{v\cdot}^{(l)}||_2 \cdot ||(\mathbf{A}\mathbf{X}^{(l)})_{v\cdot}||_2}, v = 1, ..., N,$$

Cosine Similarity of the Neighbors and the Node

$$\mathbf{s}^{(l)} = (S_1^{(l)}, ..., S_N^{(l)}).$$

All Node S values for layer l



Adaptive Discrepancy Retaining (ADR)

☐ **Shallow** layers **distill** the NDR **to** the **deep layers**

$$\mathcal{L}_{\mathrm{N}} = \sum_{l=l^\star,\ldots,L-1} \mathbb{1}(\|\mathbf{s}^{(l)}\| > \|\mathbf{s}^{(l+1)}\|) d^2(\mathbf{s}^{(l+1)}, \mathbf{s}^{(l)})$$

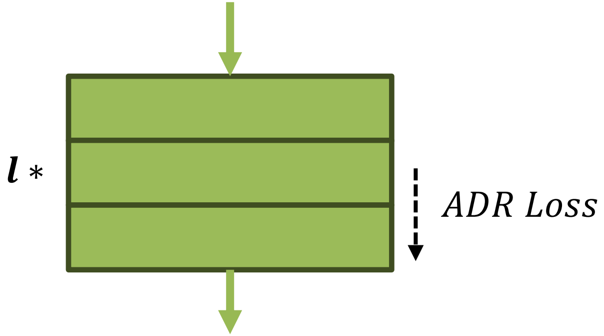
Loss Function of ADR Gate

$$l^\star = \mathrm{argmax}_k \{\|\mathbf{s}^{(k)}\| | k \in \{1, \ldots, L-1\}\}$$

Initial Layer to Start Computing

$$d^2(\mathbf{s}^{(l+1)}, \mathbf{s}^{(l)}) = \|\mathbf{D}(\mathbf{s}^{(l+1)} - \mathrm{SG}(\mathbf{s}^{(l)}))^{\mathrm{T}}\|_2^2$$

Distillation Between Current and Previous Layers



Graph-Level Loss

- Provides a **global view** of the embedded graph

$$\mathcal{C}(\mathcal{G}, \mathcal{P}) := \mathbf{G}^{(P)} = \mathrm{Readout}_{v \in \mathcal{G}}(\mathbf{X}_{v \cdot}^{(P)})$$

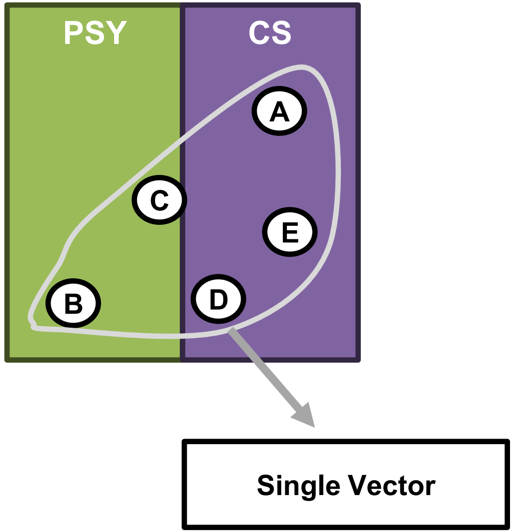
Aggregate Embedded Nodes and Make a Single Embedding

$$L_{\mathrm{G}} = \sum_{l=1,\ldots,L-1} ||\mathbf{G}^{(l+1)} - \mathrm{SG}(\mathbf{G}^{(l)})||_2^2$$

Distill Between Layers

$$\mathcal{L}_{\mathrm{T}} = \mathrm{CE}(\hat{y}, y) + \alpha\mathcal{L}_{\mathrm{L}} + \beta\mathcal{L}_{\mathrm{N}} + \gamma\mathcal{L}_{\mathrm{G}}$$

Final Loss Function



Experiments – Performance Comparison

- 5 Layer GAT / PPI Dataset

Method	Knowledge Source	F1 Score	Time
Teacher (T)	/	97.6	0.85s
Baseline	/	95.7	**0.62s**
AT	$T(\mathbf{X})$	95.4	1.75s
FitNet	$T(\mathbf{X})$	95.6	1.99s
LSP	$T(\mathbf{X})$,$T(\mathbf{A})$	96.1	1.90s
Baseline	/	$95.61_{\pm 0.20}$	**0.62s**
AT	**X**	$95.88_{\pm 0.25}$	0.73s
FitNet	**X**	$95.60_{\pm 0.17}$	0.95s
BYOT	**X**	$95.81_{\pm 0.56}$	0.80s
GNN-SD	**X,A**	$\mathbf{96.20}_{\pm 0.03}$	0.87s

Experiments – Varying-Depths Models

- Performance degradation is not seen in even the deeper layers

Dataset	Model	Layers 2	4	8	16
Cora	GAT	83.2	80.1	76.9	74.8
	GAT w/ GNN-SD	**83.7**	81.2	**80.1**	77.6
	GraphSage	81.3	80.3	78.8	77.2
	GraphSage w/ GNN-SD	81.7	**81.5**	79.8	**78.2**
Citeseer	GAT	72.5	70.5	65.1	64.5
	GAT w/ GNN-SD	72.6	**71.5**	**68.3**	**66.2**
	GraphSage	72.3	70.7	61.7	59.2
	GraphSage w/ GNN-SD	**72.7**	71.0	64.5	61.8
Pubmed	GAT	79.2	78.5	76.6	75.6
	GAT w/ GNN-SD	**79.5**	**79.4**	**78.5**	76.6
	GraphSage	78.8	77.9	73.8	77.2
	GraphSage w/ GNN-SD	79.2	79.4	77.6	**78.2**

Experiments – Ablation

	GNN-*B*	GNN-*L*	GNN-*N*	GNN-*G*	GNN-*M*	
$\mathcal{L}_{\mathrm{L}}$		√			√	√
$\mathcal{L}_{\mathrm{N}}$			√		√	√
$\mathcal{L}_{\mathrm{G}}$				√		√
Cora	80.12	79.83	80.83	/	**81.16**	/
Citeseer	70.53	71.16	71.43	/	**71.52**	/
Pubmed	78.52	78.53	**79.44**	/	79.26	/
ENZYMES	66.00	66.66	67.66	67.66	66.66	**69.33**
DD	73.00	73.76	**74.77**	73.51	73.76	74.14

Experiments – Comparison of Studies

Table 7: The effect of the adaptive self-distillation.

OGB-arxiv	GCN	GraphSage	Cora	GCN	GraphSage
baseline	$71.84_{\pm 0.18}$	$71.12_{\pm 0.38}$	baseline	$80.67_{\pm 1.14}$	$79.28_{\pm 1.91}$
naive matching	$72.08_{\pm 0.20}$	$71.62_{\pm 0.32}$	naive matching	$82.70_{\pm 0.97}$	$79.97_{\pm 1.81}$
adaptive matching	$72.34_{\pm 0.32}$	$71.88_{\pm 0.23}$	adaptive matching	$82.92_{\pm 0.89}$	$80.31_{\pm 2.35}$

Table 8: Settings of the baseline backbone used on the PPI dataset for comparing with LSP.

Model	Layers	Attention heads	Hidden features	#Params
Teacher (T)	3	4,4,6	256,256,121	3.64M
Baseline	5	2,2,2,2,2	68,68,68,68,121	0.16M

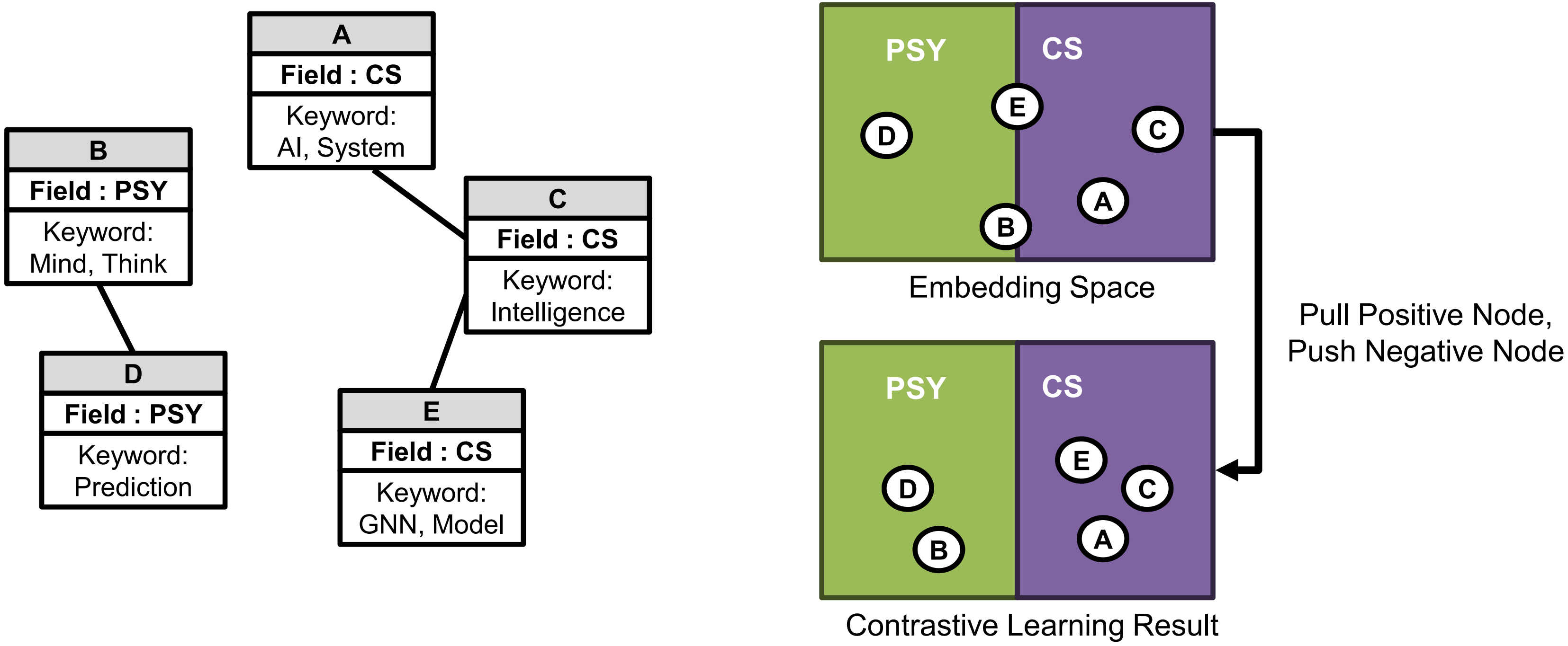
SAIL: Self-Augmented Graph Contrastive Learning

AAAI 24

Lu Yu[2,1], Shichao Pei[1], Lizhong Ding[5], Jun Zhou[2], Longfei Li[2], Chuxu Zhang[4*], Xiangliang Zhang[3,1*]
[1]King Abdullah University of Science and Technology, [2]Ant Group, [3]University of Notre Dame, [4]Brandeis University, [5]Inception Institute of Artificial Intelligence

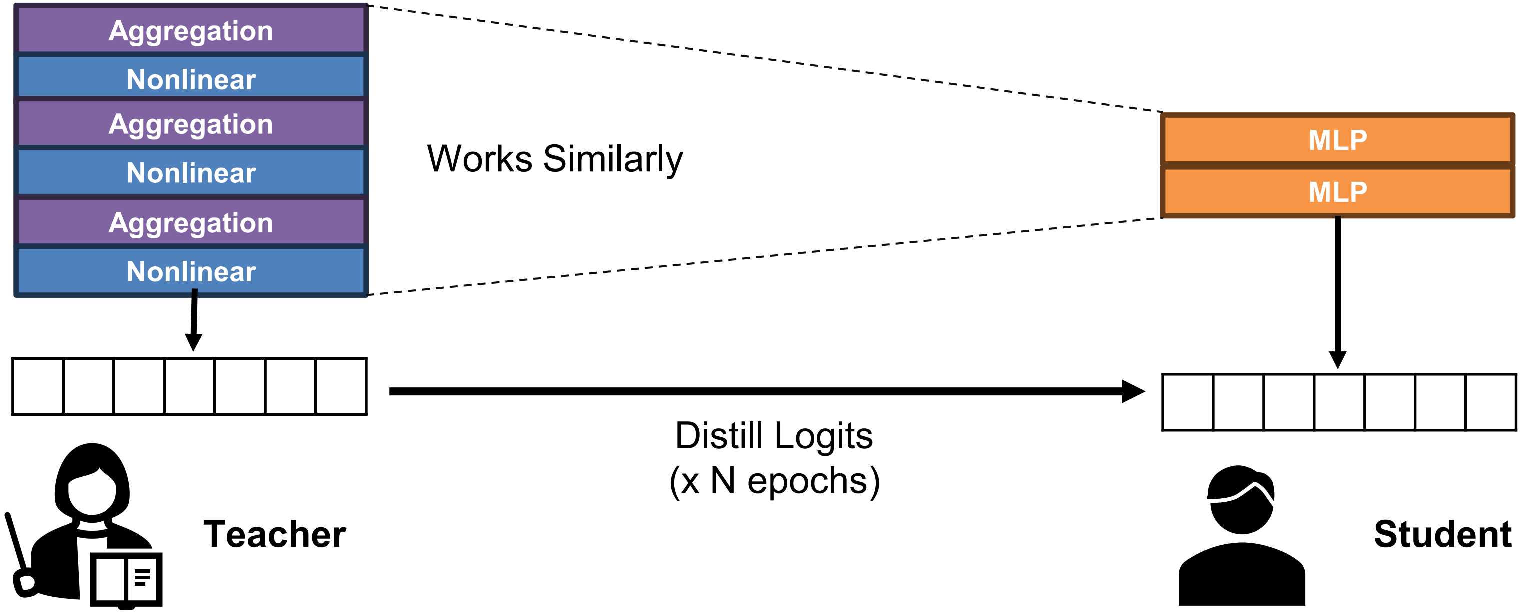
Contrastive Learning

- Contrastive learning **pulls positive** nodes and **push negative** nodes



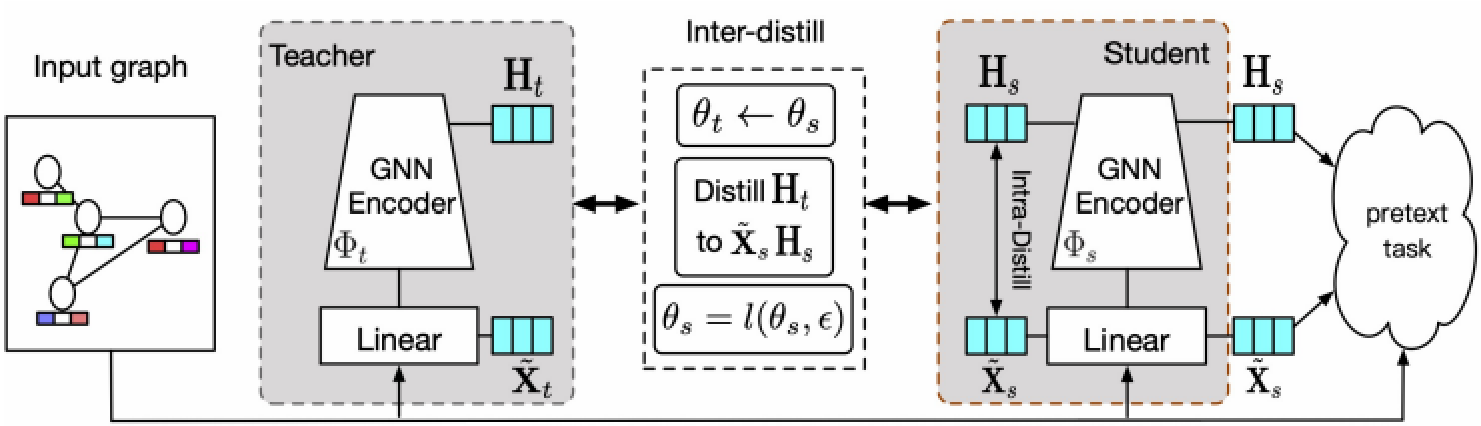
GNN to MLP Distillation

- The light **MLP** model **captures how the GNN works** and works in a similar way as the teacher model does.



SAIL Framework

- **Inter** and **intra**-distill methods are used along with **contrastive learning**
  - Used for self-supervised learning



$$\Phi(\mathbf{X}, \tilde{\mathbf{A}}) = \sigma(\tilde{\mathbf{A}}^2 \mathbf{X}\mathbf{W})$$

Loss Function Using Contrastive Learning

- ☐ Self-supervised method to make the **output learn** how to **distinguish neighbor** nodes

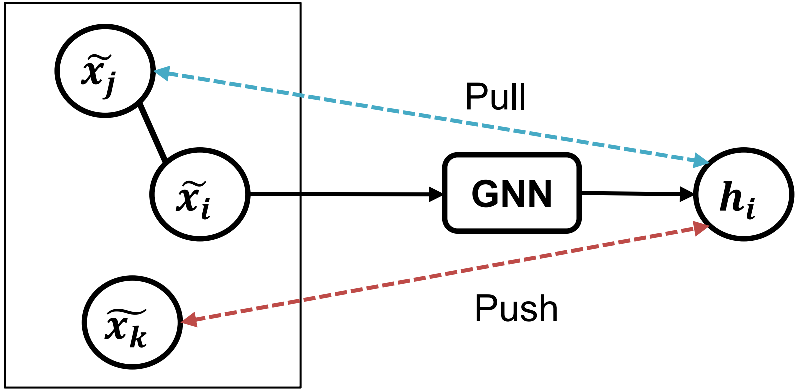
$$\mathcal{L}_{ssl} = \sum_{e_{ij}\in\mathcal{E}} \sum_{e_{ik}\notin\mathcal{E}} -\ell(\psi(h_i, \widetilde{x}_j), \psi(h_i, \widetilde{x}_k)) + \lambda\mathcal{R}(\mathcal{G})$$

Loss Function of the Framework

$$\ln\sigma(\psi(h_i, \widetilde{x}_j) - \psi(h_i, \widetilde{x}_k))$$

$$\sigma(x) = \frac{1}{1+exp(-x)}$$

Contrastive Loss l



Intra-distilling Module

- **Distill distance** between **center** node and its **sampled pseudo** local neighbors at a subgraph-level

$$S^t_{ij} = \frac{exp(\psi(h_i, \widetilde{x}_j))}{\sum_{v_j \in LS_i} exp(\psi(h_i, \widetilde{x}_j))}$$

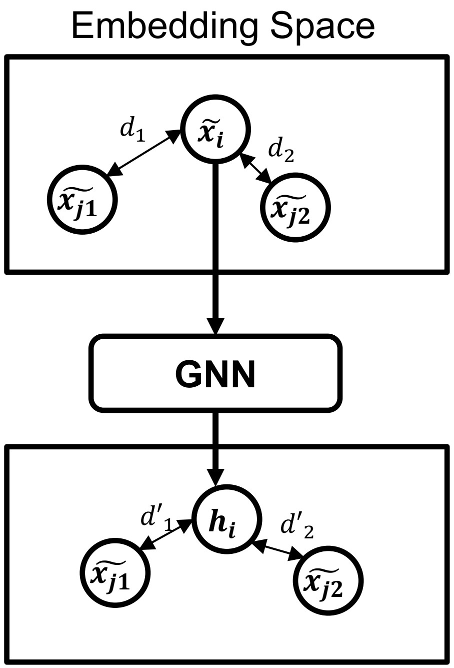
$$S^s_{ij} = \frac{exp(\psi(\widetilde{x}_i, \widetilde{x}_j))}{\sum_{v_j \in LS_i} exp(\psi(\widetilde{x}_i, \widetilde{x}_j))}$$

Preserve Embedding Distance Between Nodes

$$\mathcal{S}_i = CrossEntropy(S^t_{[i,\cdot]}, S^s_{[i,\cdot]})$$

$$\mathcal{R}_{intra} = \sum_{i=1}^{N} \mathcal{S}_i$$

Distillation



Inter-distilling Module

- Make a noisy teacher, and train by repeatedly imitating the teacher

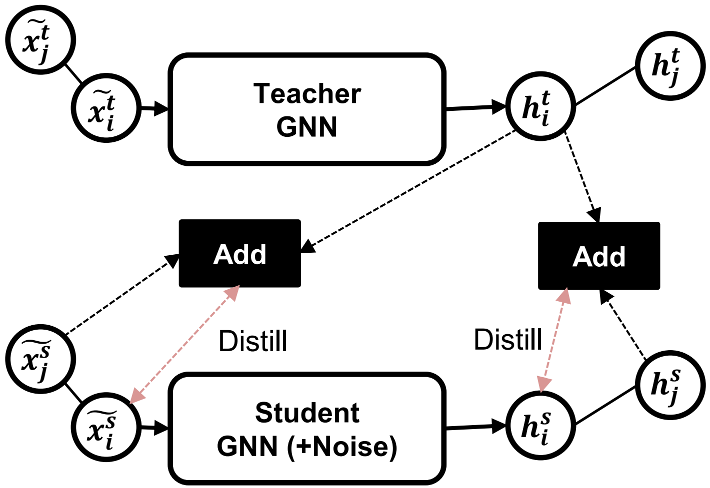
$$KD(\mathbf{H}_t, \widetilde{\mathbf{X}}_s|\mathcal{G}) = ||\widetilde{x}_i^s - \frac{(1-\alpha)h_i^t + \alpha\sum_{j\in\mathcal{N}(i)}\beta_{ij}\widetilde{x}_j^s}{(1-\alpha)+\alpha\sum_{j\in\mathcal{N}(i)}\beta_{ij}}||_2^2$$

$$KD(\mathbf{H}_t, \mathbf{H}_s|\mathcal{G}) = ||h_i^s - \frac{(1-\alpha)h_i^t + \alpha\sum_{j\in\mathcal{N}(i)}\beta_{ij}h_j^s}{(1-\alpha)+\alpha\sum_{j\in\mathcal{N}(i)}\beta_{ij}}||_2^2$$

Imitate 2-hop GNN During Distillation

$$\mathcal{R}_{inter} = KD(\mathbf{H}_t, \widetilde{\mathbf{X}}_s|\mathcal{G}) + KD(\mathbf{H}_t, \mathbf{H}_s|\mathcal{G})$$

Inter-distill Module



Experiment – Node Classification

☐ 1-Layer GCN model in an unsupervised setting

Input	Methods	Cora	Citeseer	PubMed	Computers	Photo	CS
X,A,Y	ChebNet (Defferrard, Bresson, and Vandergheynst 2016)	81.2	69.8	74.4	70.5±0.5	76.9±0.3	92.3±0.1
	MLP (Kipf and Welling 2017)	55.1	46.5	71.4	55.3±0.2	71.1±0.1	85.5±0.1
	GCN (Kipf and Welling 2017)	81.5	70.3	79.0	76.11±0.1	89.0±0.3	92.4±0.1
	SGC (Wu et al. 2019)	81.0 ± 0.0	71.9±0.1	78.9±0.0	55.7±0.3	69.7±0.5	92.3±0.1
	GAT (Veličković et al. 2018)	83.0±0.7	72.5±0.7	79.0±0.3	71.4±0.6	89.4±0.5	92.2±0.2
	DisenGCN (Ma et al. 2019)	83.7*	73.4*	80.5	52.7±0.3	87.7±0.6	93.3±0.1*
	GMNN (Qu, Bengio, and Tang 2019)	83.7*	73.1	81.8*	74.1±0.4	89.5±0.6	92.7±0.2
	GraphSAGE (Hamilton, Ying, and Leskovec 2017)	77.2±0.3	67.8±0.3	77.5 ±0.5	78.9±0.4	91.3±0.5	93.2 ±0.3
X,A	CAN (Meng et al. 2019)	75.2±0.5	64.5±0.2	64.8±0.3	78.2±0.5	88.3±0.6	91.2±0.2
	DGI (Veličković et al. 2019)	82.3±0.6	71.8±0.7	76.8±0.6	68.2±0.6	78.2±0.3	92.4±0.3
	GMI (Peng et al. 2020)	82.8±0.7	73.0±0.3	80.1±0.2	62.7±0.5	86.2±0.3	92.3±0.1
	MVGRL (Hassani and Khasahmadi 2020)	83.5±0.4	72.1±0.6	79.8±0.7	88.4±0.3*	**92.8**±0.2	93.1±0.3
	SAIL	**84.6**±0.3	**74.2**±0.4	**83.8**±0.1	**89.4**±0.1	92.5±0.1*	**93.3**±0.05

Table 1: Accuracy (with standard deviation) of node classification (in %). The best results are highlighted in bold. Results without standard deviation are copied from original works.

Experiment – Mean Average Distance

- Large MAD ratio values show that **the over-smoothing occur less**

Data	Metrics	GCN	GAT	DisenGCN	GMNN	CAN	DGI	GMI	MVGRL	SAIL
Cora	MAD_{nei}	0.075	0.029	0.215	0.088	0.059	0.312	0.069	0.240	0.013
	MAD_{gap}	0.308	0.083	0.471	0.390	0.900	0.557	0.966	0.661	0.322
	MAD_{ratio}	4.13	2.86	2.19	4.43	15.2	1.77	13.83	2.75	25.3
Citeseer	MAD_{nei}	0.049	0.014	0.194	0.059	0.057	0.289	0.122	0.174	0.007
	MAD_{gap}	0.308	0.083	0.471	0.390	0.921	0.497	0.879	0.491	0.429
	MAD_{ratio}	4.13	2.86	2.19	4.43	15.1	1.72	6.2	2.82	61.2
Pubmed	MAD_{nei}	0.043	0.024	0.054	0.068	0.037	0.112	0.086	0.180	0.024
	MAD_{gap}	0.155	0.083	0.224	0.438	0.757	0.145	0.833	0.541	0.294
	MAD_{ratio}	6.02	6.43	4.17	6.39	20.6	1.29	8.63	3.00	12.1

Table 5: Empirical analysis to shown the quality of learned node representations measured by mean average distance (MAD).

Experiment – Ablation Study

- Node classification accuracy

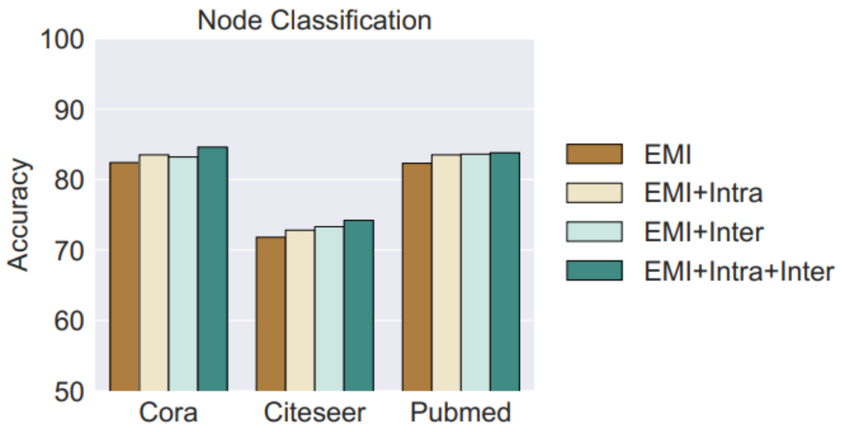
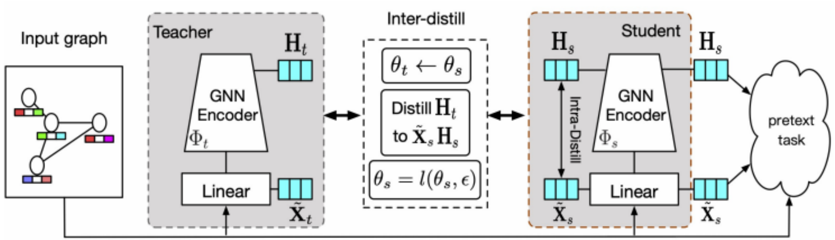
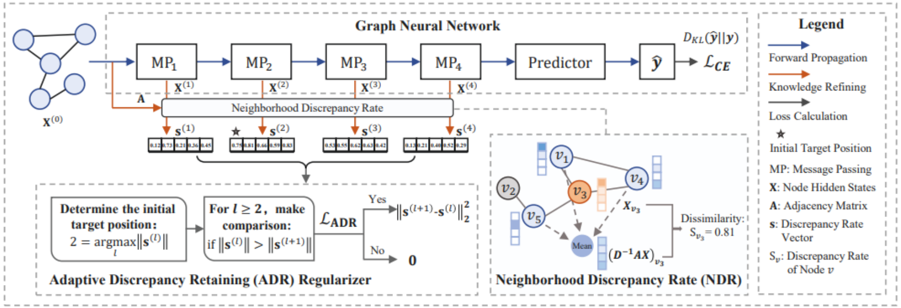


Figure 3: Ablation study of the distillation component influence in node classification accuracy (in %).

Conclusion

- Over-smoothing is an issue that should be dealt with when using deep layers in GNN
- Self-distillation can be a key solution for over-smoothing
- GNN-SD uses the similarity between neighbors to guess the smoothness and fix by distilling previous layers to over-smoothened layers
- SAIL learns by imitating the deep-layer GCN while only using a single layer, reducing over-smoothing while preserving the performance





Thank You!